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Palaeomagnetism of alkaline lava flows from El Salto-Almafuerte, Córdoba Province, Argentina

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Summary. Stable components of magnetization have been isolated in 15 lava flows (mean K–Ar age 123 ± 4 Myr) from the alkaline sequence outcropping at El Salto-Almafuerte, Province of Córdoba, Argentina. Magnetic and geologic stratigraphy, as well as K–Ar ages indicate that this sequence was probably extruded in the Lower Cretaceous during the first volcanic cycle of the Sierra de los Cóndores Group (Vulcanitas Cerro Colorado Formation).

The palaeomagnetic pole-position for El Salto-Almafuerte lava flows, computed from the mean of 15 virtual geomagnetic poles and denoted SAK_7, is: 25° E, 72° S ($k = 35$, $\alpha_{95} = 6.5°$); it is fairly close to other Lower Cretaceous palaeomagnetic poles for South America. The elongated distribution of Cretaceous palaeomagnetic poles suggest recurrent drift for South America in early Cretaceous time.

The palaeomagnetic and radiometric data for the igneous rocks from El Salto-Almafuerte support the magnetic reversal time-scale for the early Cretaceous suggested by oceanic magnetic lineations.

1 Introduction

When South America and Africa are joined at their continental shelves the palaeomagnetic poles computed from the stable magnetic remanence of Late Palaeozoic and Early Mesozoic rocks of these continental blocks overlapped (Valencio & Vilas 1972). On this palaeogeographic reconstruction the palaeomagnetic pole for the Serra Geral Formation (Creer 1962) does not coincide with the contemporaneous African palaeomagnetic poles, suggesting a pre-Lower Cretaceous age for the opening of the South Atlantic Ocean. This interpretation was supported by only one South American palaeomagnetic pole and Valencio (1972) emphasized the importance of performing more palaeomagnetic studies on South American Cretaceous formations.

In the southern part of Córdoba province, Argentina, Lower Cretaceous igneous and sedimentary rocks of continental origin outcrop in several localities. Gordillo & Lencinas (1967) have included these rocks in the Sierra de los Cóndores Group. Palaeomagnetic

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studies of different exposures of the Sierra de los Cóndores Group have been reported by Valencio (1972), Linares & Valencio (1975) and Vilas (1976). In this paper, results of the palaeomagnetic study of the vulcanites of the same Group, exposed at El Salto-Almafuerte, are given.

2 Geological setting

The igneous and sedimentary sequence studied here outcrops at the margins of the Rio Tercero river at El Salto, two kilometres north of Almafuerte, in the southern part of Córdoba province, Argentina (32° 10′ S, 64° 15′ W), where a volcanic sequence of alkaline–olivine basalts with a few continental sedimentary intercalations overlie granitic basement (Fig. 1, Schröder 1967); the strike and dip of the layers is uniform in the whole area (strike NNW, dip 25–30° to the east). The thickness of the volcanic-sedimentary sequence is about 160 m.

Gordillo & Lencinas (1972) correlate outcrops at El Salto-Almafuerte with those of the Sierra de los Cóndores Group. According to these authors (Gordillo & Lencinas 1967) the Sierra de los Cóndores Group consists of four formations which, in chronological order from the base to the top are: Embalse Río Tercero Formation (sediments, mainly red beds),

Table 1. K–Ar isotopic analyses of two samples from El Salto-Almafuerte upper flow.

Sample No.	K (%)	Ar^{40}_{rad} (mol/gm)	Ar^{40}_{atm} (%)	Isotopic age (my)	Mean age (my)
3373 Whole rock	4.14	9.009×10^{-10}	2.0	119±10	
3375 Whole rock	4.20	9.384×10^{-10}	4.6	121±5	120±2

Vulcanitas Cerro Colorado Formation (lava flows with interbedded continental sediments), Conglomerado Cerro Libertad Formation (polimictic orthoconglomerate) and Vulcanitas Rumipalla Formation (lava flows). Samples from the first volcanic cycle (Vulcanitas Cerro Colorado Formation) were dated by the K–Ar method giving ages from 112 to 128 Myr (six determinations) with a mean of 121 ± 3 Myr (Gordillo & Lencinas 1967; González & Tosselli 1978). The last quoted authors also give K–Ar ages for a dyke intruding the granitic basement (120 ± 2 Myr) and for two lava flows of the lower part of the sequence at El Salto-Almafuerte (129 ± 8 and 122 ± 3 Myr).

K–Ar age determinations for the upper flow of this sequence were made on two of the palaeomagnetic samples (Table 1). The determinations were carried out by E. Linares in the radiometric laboratories of the INGEIS at the University of Buenos Aires, Argentina. Techniques of measurements were those described by Linares, Kawashita & Kleiner (1973). The ages obtained were 119 ± 10 and 121 ± 5 Myr; then the mean K–Ar age for the upper flow at El Salto-Almafuerte is 120 ± 2 Myr (Table 1). The mean age for the four dated geologic units of this locality is 123 ± 4 Myr. This mean age and the presence of interbedded sediments suggests that the vulcanites studied here correspond to the first volcanic cycle of the Sierra de los Cóndores Group (i.e. the Vulcanitas Cerro Colorado Formation).

Lava flows belonging to the two volcanic cycles of the Sierra de los Cóndores Group were studied by Valencio (1972), Cerro Colorado area, first volcanic cycle and Vilas (1976), El Estrecho-Cerro Libertad area, second volcanic cycle. Here we present the results of the palaeomagnetic study carried out with alkaline lava flows extruded during the first volcanic cycle at El Salto-Almafuerte area, 20 km to the east of the Cerro Colorado area.

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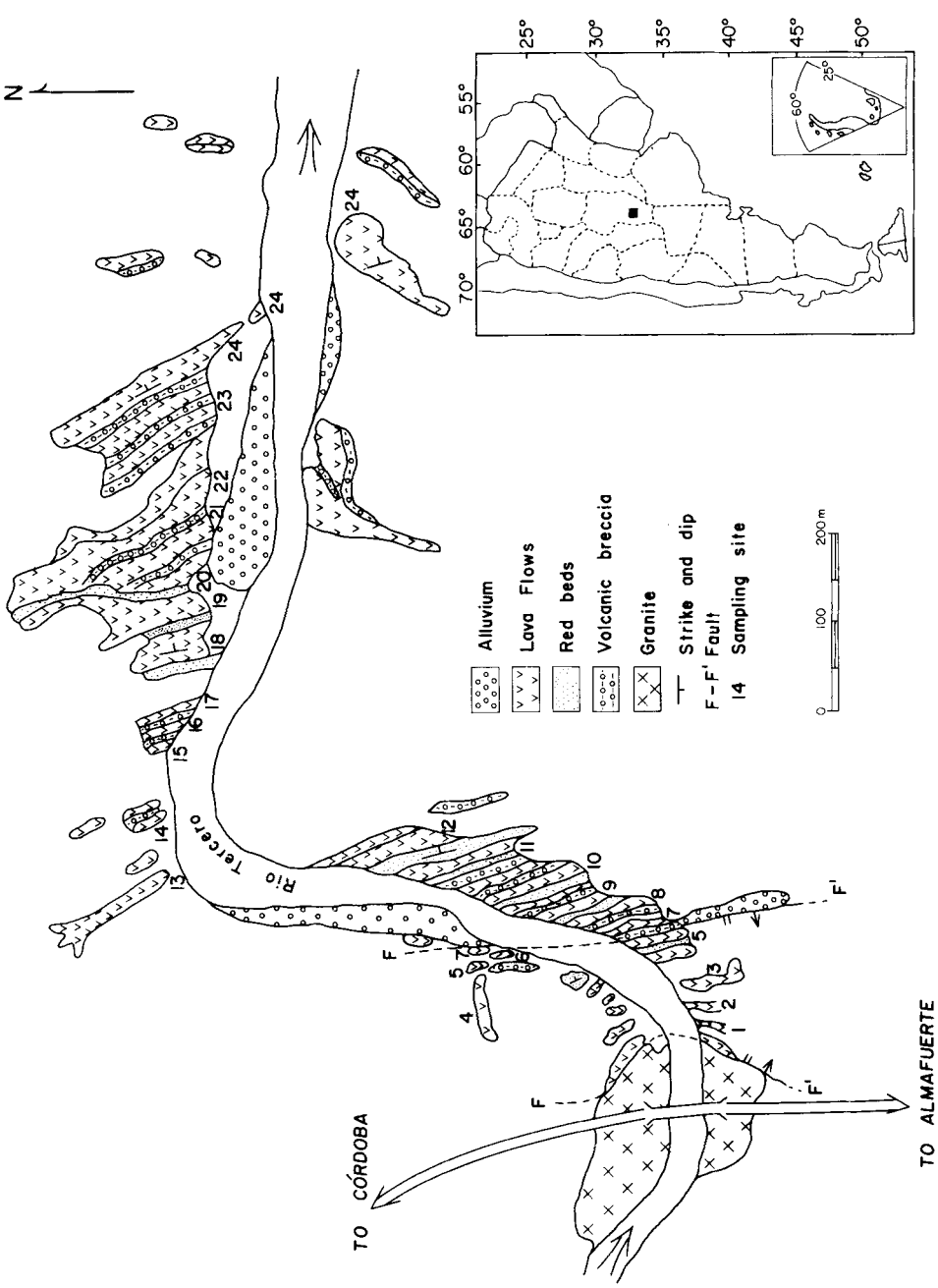


Figure 1. Geological scheme at El Salto-Almafuerte area (Province of Córdoba, Argentina) modified after Schröder (1967). The units included in the palaeomagnetic study are indicated by numbers.

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Table 2. Summary of the palaeomagnetic data for the units of Vulcanitas Cerro Colorado Formation from El Salto-Almafuerte, Province of Córdoba, Argentina.

Rock unit	Number of samples used	Directions of cleaned NRM referred to the palaeohorizontal						Virtual geomagnetic poles					
		N	D (°)	I (°)	α_{95}	k	Polarity	Name	Lat (°) (south)	Long (°) (east)	$d\psi$ (°)	$d\chi$ (°)	Symbol
24	12	10	159	42	5	97	R	24	70.2	43.3	3.7	6.1	◢
23	2	1	134	44	–	–	R						
22	6	2	172	45	10	588	R	22	81.1	66.7	8.2	13.0	◤
21	9	5	163	45	12	40	R	21	74.6	42.2	9.9	15.0	◣
20	5	3	169	56	8	156	R	20	79.9	356.6	8.2	11.4	◢
19	3	–					–						
18	4	3	335	−54	9	177	N	18	69.3	13.9	9.2	13.0	▷
17	3	2	331	−69	14	316	N	17	60.7	335.6	20.3	23.9	◇
16	3	–					–						
15	2	2	153	52	9	808	R	15	67.1	19.7	8.2	12.0	◡
14	2	–					–						
13	5	3	151	53	12	99	R	13	65.8	17.1	11.8	17.2	▼
12	4	–					–						
11	3	2	160	47n	11	503	R	11	70.5	34.0	9.3	14.4	◆
10	8	6	325	−54	6	144	N	10	60.8	13.3	5.5	7.9	△
9	3	–					–						
8	4	2	176	34	21	143	R	8	76.3	100.0	14.1	24.7	●
7	15	10	313	−48	4	170	N	7	49.6	17.5	3.4	5.2	□
6	4	4	162	52	12	60	R	6	74.4	23.0	11.1	16.3	■
5	10	5	182	51	4	297	R	5	88.5	201.0	3.7	5.4	▲
4	6	4	105	31	7	169	R	4	21.2	20.9	4.4	7.9	⬮
3	3	1	130	44	–	–	R						
2	5	–					–						
1	2	2	331	−30	3	9362	N	1	59.2	52.3	1.6	2.9	○

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3 Palaeomagnetic study

Sampling was carried out by J. F. A. Vilas, B. J. J. Embleton and D. A. Valencio on the basis of the detailed geological study made by Schröder (1967) (Fig. 1). One hundred and twenty-three oriented hand samples were collected (Table 2). The orientation of samples was carried out using both magnetic and Sun compass. In most cases Sun and magnetic azimuths agreed within 6° or less, after correction for magnetic declination, so the average of both measurements was taken for field correction. Two or three cylinders, 2.5 cm in diameter, were cut from each hand sample; then four disks 1-cm thick were cut from these cylinders. Natural remanent magnetization (NRM) of these disks was measured using spinner and fluxgate magnetometers.

One disk of each sample was subjected to AC demagnetizing fields increasing in steps of 50 Oe up to 300 Oe peak field and the best demagnetizing field strength was chosen as that beyond which the direction of the remanence of the pilot disk showed no further change to higher demagnetizing field strengths. Optimum values varied from 100 to 300 Oe. A second and sometimes a third disk of the same sample was submitted to the selected AC demagnetizing field in order to obtain a better definition of the stable remanence of the sample.

In some cases the remanent magnetization of the pilot sample was sharply reduced after treatment in AC fields of 50–100 Oe; for those samples the AC cleaning was done in steps of 25 Oe up to 200 Oe peak field. These samples after the step of 200 Oe kept less than 0.2 of their NRM intensities ($J_r/J_{r_0} < 0.2$). Highly-scattered directions of NRM were often found associated with these soft components, notably in lava flows 13, 15, 17, 20, 21 and 22.

The mean direction of stable remanence of each unit was calculated using Fisher's (1953) statistics (Table 2). Within-site precision for the total NRM is good for the units 1, 4, 6, 7, 11 and 18, with k values as high as 2810 for unit 1, but for other units it varies from moderate to poor. On the other hand directions of the stable cleaned remanence of samples from each of the units are well grouped indicating that AC cleaning has eliminated secondary magnetizations.

Many samples, notably those of units 2, 9, 12, 14, 16 and 19, show unstable NRM, probably due to water weathering in the lower levels of outcrops at both margins of the river. Units 3 and 23 have only one sample with stable remanent magnetization. The mean directions of the stable remanent magnetization of each of the remaining units, after dip correction, are given in Table 2 and Fig. 2. Virtual geomagnetic poles for each one of these geological units are also given in Table 2. Unit 4 yields a mean direction removed from those of the other units and could have been extruded during a reversal of the geomagnetic field; for that reason it is not used in estimating the palaeomagnetic pole position for the Vulcanitas Cerro Colorado Formation at El Salto-Almafuerte. Fig. 3 shows the virtual geomagnetic poles (squares) computed for this formation (except unit 4) together with the South American Cretaceous geomagnetic poles (SAK_1, Creer (1962); SAK_2, Valencio (1972); SAK_3, Linares & Valencio (1975); SAK_4, Vilas (1976); SAK_5, Opdyke & MacDonald (1973) and SAK_6, Valencio *et al.* (1975)).

The reversals of the geomagnetic field recorded in the lava flows from El Salto-Almafuerte and the number of time units involved, suggest that the mean of these virtual geomagnetic poles (SAK_7, 25° E, 72° S, $\alpha_{95} = 6.5$, $k = 35$) may be considered as a palaeomagnetic pole. Its position is reasonably close to palaeomagnetic poles for the Lower Cretaceous Serra Geral Formation (SAK_1), Vulcanitas Cerro Colorado Formation (SAK_2) and Trachybasaltic dykes from Río Los Molinos (SAK_3). In particular there is the possibility that the time embraced by the volcanic episode studied here overlaps the period of extrusion of the lavas from the Cerro Colorado area (SAK_2, Valencio 1972). However, all the igneous and sedimentary rocks collected at Cerro Colorado and included in that study have reversed

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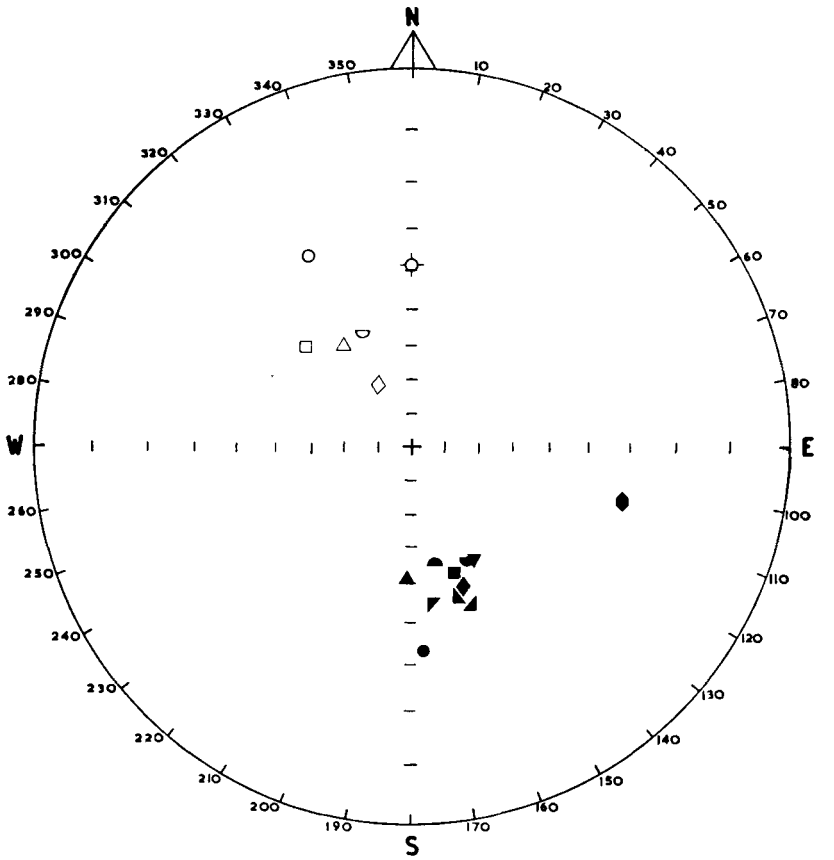


Figure 2. Mean directions of stable remanent magnetization, as referred to the palaeohorizontal, for the geological units studied (Table 1). The direction of the axial dipole is also shown by ⊹.

magnetization. On the other hand the position of SAK_7 differs from the palaeomagnetic pole computed for the Vulcanitas Rumipalla Formation (SAK_4, Vilas 1976). The age of this formation is younger than that of the Cerro Colorado Formation. Fig. 3 shows that the position of SAK_7 also differs from those of the palaeomagnetic poles for the Lower–Upper Cretaceous Pirgua Subgroup (SAK_6) and the Upper Cretaceous Poço de Caldas Alkaline Complex (SAK_5). The elongated distribution of the South American Cretaceous palaeomagnetic poles from east to west accords with their relative ages (Fig. 3) and suggests, as was indicated by Valencio *et al.* (1975), an episode of recurrent drift of the South American block during the Cretaceous period.

4 Cretaceous palaeomagnetic polarity column

Valencio (1972) made a critical analysis of the Cretaceous palaeomagnetic data summarizing the available information in a palaeomagnetic polarity column for the Cretaceous (see Fig. 6 of Valencio 1972). He also suggested the existence of a zone of reverse polarity of the geomagnetic field at about 119 Myr (Sierra de los Cóndores zone). The palaeomagnetic and radiometric data from El Salto-Almafuerte volcanic sequence suggest a redefinition of the Lower Cretaceous mixed interval of that part of the polarity column. Actually, the reversed lava flows 5 and 6 at the lower part of the sequence have a mean K–Ar age of

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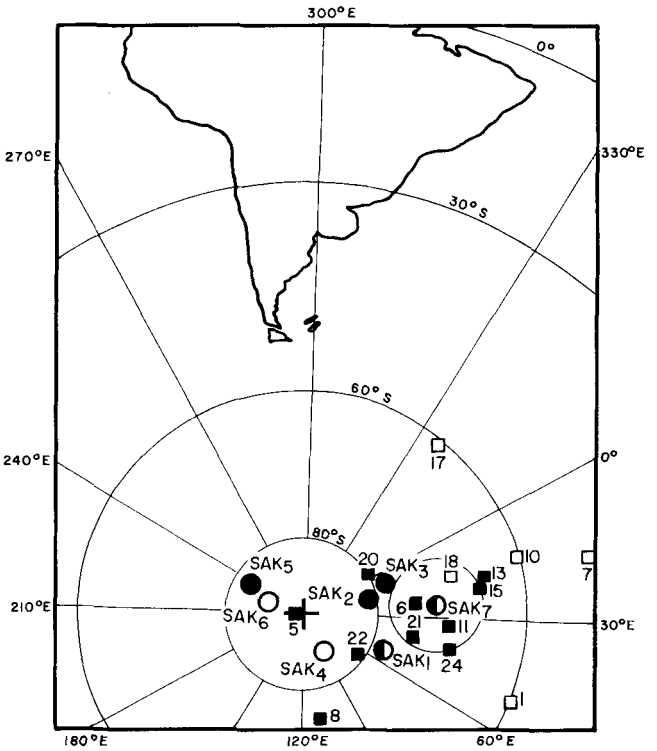


Figure 3. Virtual geomagnetic poles (□) for 15 units of Vulcanitas Cerro Colorado Formation (Table 1) from El Salto-Almafuerte and Cretaceous palaeomagnetic poles (○) for South America (references, see item 3).

125 ± 5 Myr, and six reversals of the geomagnetic field have been recorded in the lava flows which overlie them; the highest lava flow has reverse remanence and a mean age of 120 ± 2 Myr. This is in good agreement with the magnetic reversal time-scale for the Early Cretaceous and Late Jurassic proposed by Larson & Hilde (1975) from several profiles made across lineated magnetic anomalies in the Pacific Ocean.

5 Conclusions

Stable remanent magnetizations of 15 time units were used to compute a pole (SAK_7) for the Lower Cretaceous (mean age 123 ± 4 Myr) Vulcanitas Cerro Colorado Formation of the Sierra de los Cóndores Group at El Salto-Almafuerte, Province of Córdoba, Argentina. The number of geologic units employed and the seven reversals of the geomagnetic field recorded, suggest that SAK_7 is a palaeomagnetic pole. Its position is reasonably close to three other Lower Cretaceous South American poles (SAK_1, SAK_2 and SAK_3, Fig. 3). The mean of the positions of these four poles (78° S, 27° E, $k = 220$, $\alpha_{95} = 6°$) defines the early Lower Cretaceous pole position relative to South America with more precision.

The palaeomagnetic pole for the Vulcanitas Rumipalla Formation (SAK_4) is younger in age than SAK_2 and SAK_7 and its position is intermediate between the early Lower Cretaceous and Upper Cretaceous pole positions. It suggests movement relative to the poles during the Cretaceous.

The palaeomagnetic and radiometric data for continental rocks from the Vulcanitas

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Cerro Colorado Formation collected at El Salto-Almafuerte redefine the magnetic reversal time-scale for the Early Cretaceous (McElhinny & Burek 1971; Creer 1971 and Valencio 1972). They are fairly concordant with the reversal pattern suggested by Larson & Hilde (1975) from oceanic magnetic anomalies between 120 and 125 Myr ago.

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